Experiment 2.1

3D CAD Modelling of a Camera-Shaped Component

Aim

To create a 3D camera-shaped component in a CAD application by sketching the required profiles, applying dimensions, and using extrusion and other modelling features.

Requirements

• Computer with a 3D CAD software (such as AutoCAD or an equivalent solid-modelling program)
• Mouse/keyboard for sketching and modelling
• The given dimensioned reference sketch

Procedure

1. Draw the base profile and apply dimensions. Start a 2D sketch on the appropriate plane. Use the reference dimensions: overall rectangular base width 100 mm and height 60 mm; central circular profile diameter 50 mm. Add the upper features as shown, including the 30 mm central raised section, 20 mm right-side section, and 10 mm left-side section. The small vertical offsets shown are 5 mm and 10 mm. Keep the profile centered on the indicated axes where shown.

2. Sketch the main body. Draw the rectangular body outline and the circular front feature using the dimensioned sketch as a guide. Check that the circle is centered on the horizontal and vertical reference axes.

Sketch reference

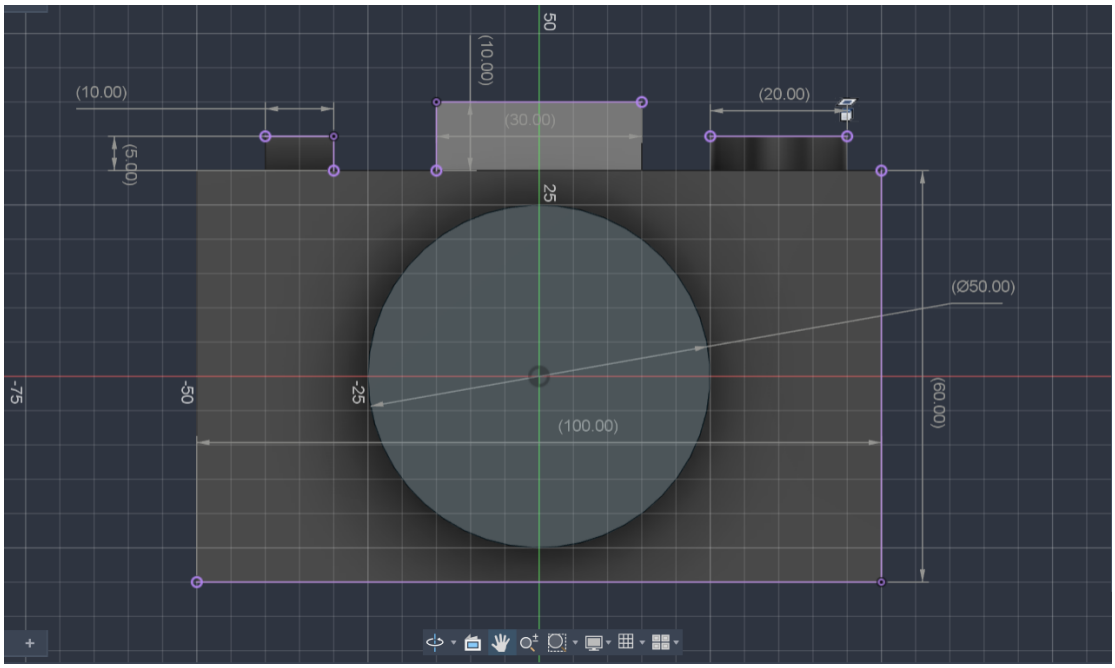


Figure 1. Dimensioned 2D sketch used as the modelling reference.

Procedure (continued)

3. Extrude the sketch. Select the closed profiles and apply the required extrusion depth to form the solid body. Extrude the circular front profile to create the projecting cylindrical lens, and form the raised top details according to the reference model.

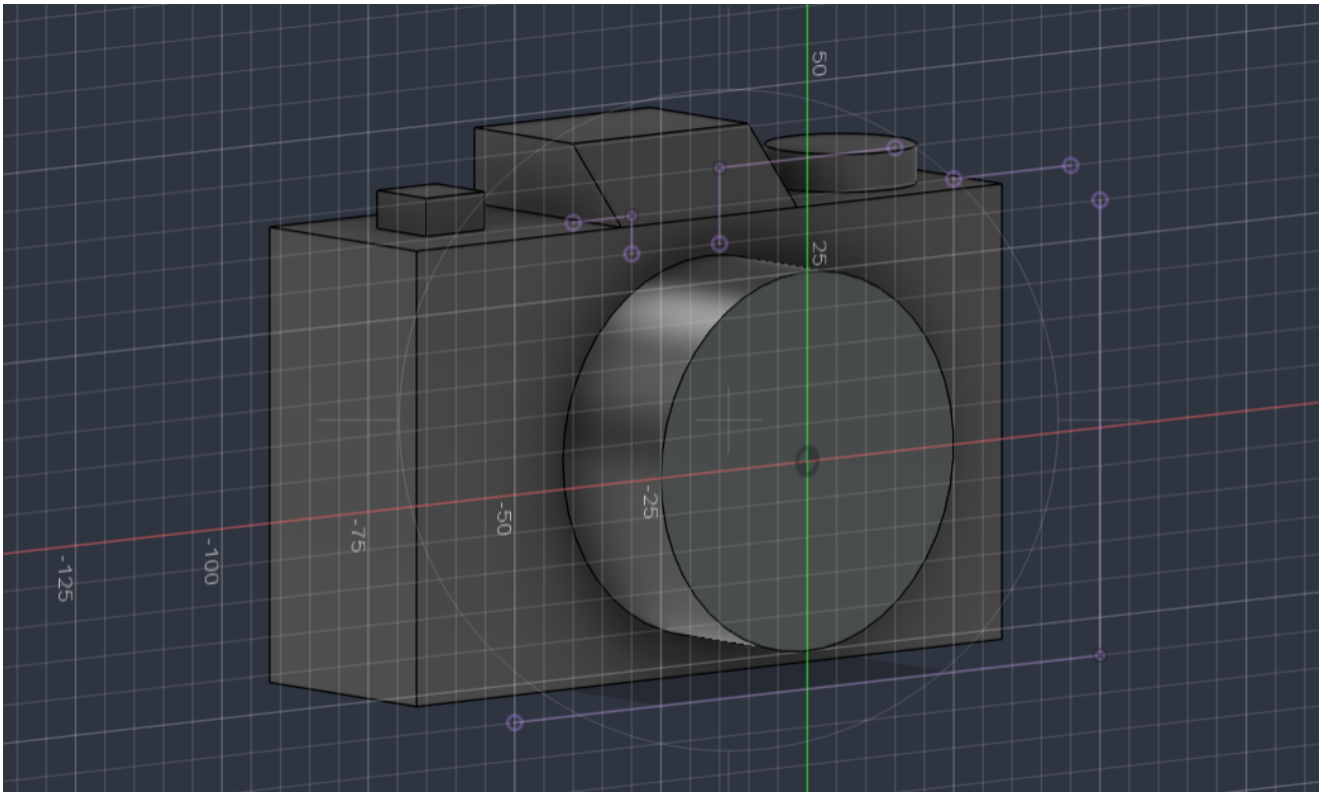


Figure 2. 3D model after extrusion and formation of the raised features.

4. Refine and inspect the model. Review the front and isometric views, confirm that the cylindrical feature and top blocks are positioned correctly, and adjust the model if necessary.

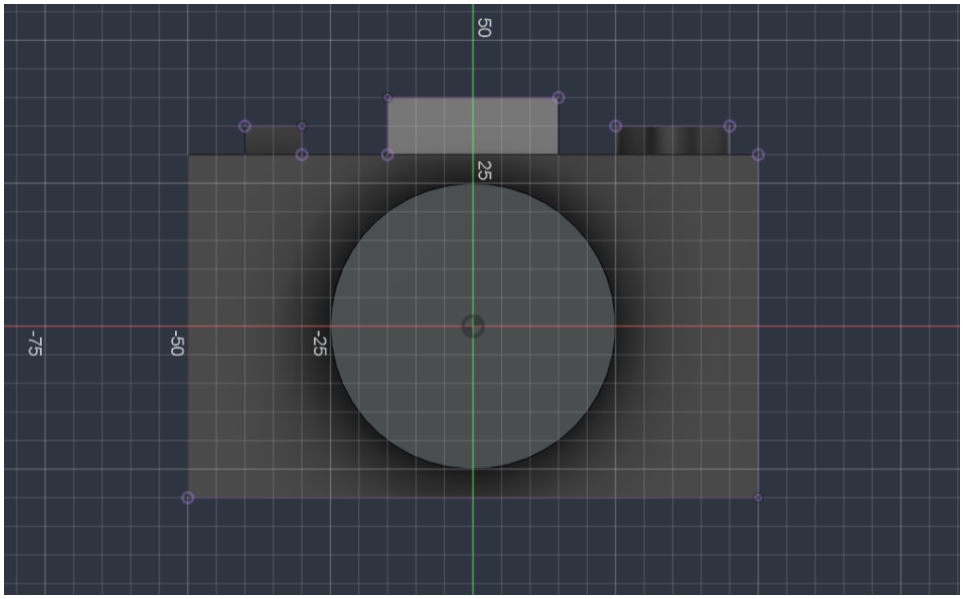


Figure 3. Front-view check of the model.

Sketch

The initial dimensioned sketch is shown in Figure 1. The 2D profile provides the outline and feature locations for the subsequent solid-modelling operations.

Output

The final output is the completed engineering drawing sheet shown below, containing isometric representations and orthographic views of the camera-shaped component.

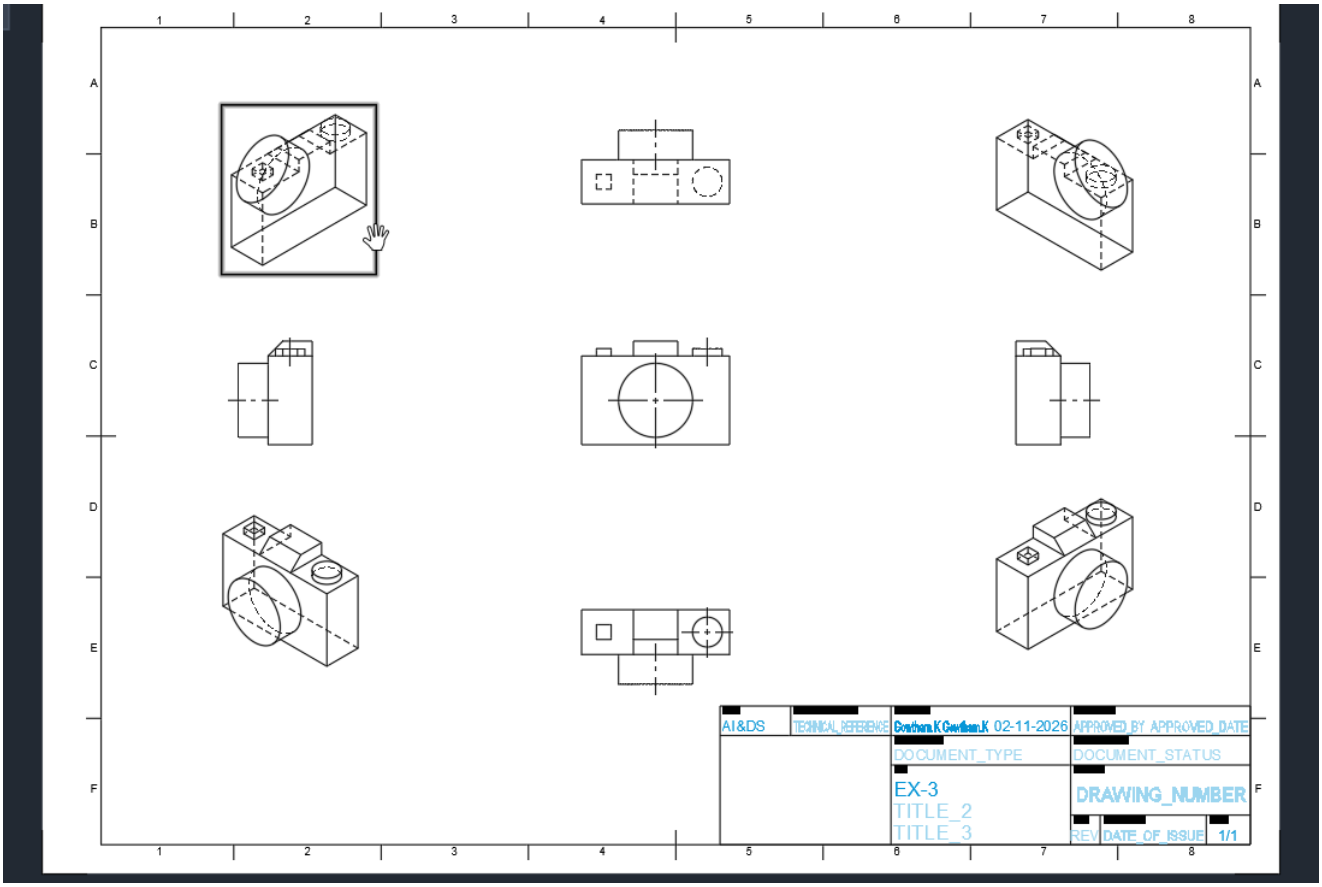


Figure 4. Final drawing sheet (last reference image) showing the completed model and its views.

Result

The camera-shaped 3D component was modelled from the given dimensions using sketching and extrusion operations. The final drawing sheet presents the resulting solid in multiple views.